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THE POSITIVE AND NEGATIVE SQUARE-ENERGY CONJECTURE

YINCHEN LIU, QUANYU TANG, AND SHENGTONG ZHANG

Abstract. Let $s^+(G)$ and $s^-(G)$ denote the sums of the squares of the positive and negative adjacency eigenvalues of a graph G, respectively. We prove the conjecture of Elphick, Farber, Goldberg, and Wocjan that every connected graph G on n vertices satisfies

$$\min\{s^+(G), s^-(G)\} \geq n-1.$$

The proof introduces a new framework for square-energy estimates, in which the Hadamard squares of positive semidefinite matrices that encode these spectral quantities are relaxed to the full doubly nonnegative cone.

1. Introduction

All graphs in this paper are finite, simple, and undirected. Let $G = (V, E)$ be a graph with $n = |V|$ vertices, $m = |E|$ edges, and adjacency matrix $A = A(G)$, and let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of A, counted with multiplicity. The *positive square energy* and *negative square energy* of G are defined by

$$s^+(G) := \sum_{\lambda_i > 0} \lambda_i^2, \qquad s^-(G) := \sum_{\lambda_i < 0} \lambda_i^2.$$

These invariants are quadratic analogues of the classical graph energy [16], and they arise naturally in spectral bounds on the chromatic number: confirming a conjecture of Wocjan and Elphick [23], Ando and Lin [5] proved that every graph with at least one edge satisfies

$$\chi(G) \geq 1 + \max\left\{ \frac{s^+(G)}{s^-(G)}, \frac{s^-(G)}{s^+(G)} \right\}.$$

Here and below, $X \succeq 0$ means that X is positive semidefinite. If

$$A = A_+ - A_-, \qquad A_+, A_- \succeq 0, \qquad A_+ A_- = 0$$

is the decomposition of A into its positive and negative spectral parts, then

$$s^+(G) = \operatorname{tr}(A_+^2), \qquad s^-(G) = \operatorname{tr}(A_-^2).$$

Since $\operatorname{tr}(A^2) = 2m$, we have

$$s^+(G) + s^-(G) = 2m. \tag{1}$$

In 2016, Elphick, Farber, Goldberg, and Wocjan [11, Conjecture 1] proposed the following conjecture.

Conjecture 1.1. *Every connected graph G on n vertices satisfies*

$$\min\{s^+(G), s^-(G)\} \geq n-1.$$

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This conjecture was later highlighted as Conjecture 1 in the survey of Liu and Ning [17].

By (1), the conjecture states, equivalently, that every connected graph satisfies $\max\{s^+, s^-\} \le 2m - n + 1$, strengthening Hong's classical spectral radius bound $\lambda_1^2 \le 2m - n + 1$ [14]. The conjectured bound is sharp: every tree satisfies $s^+(G) = s^-(G) = m = n - 1$, and the complete graph satisfies $s^-(K_n) = n - 1$.

Since its formulation, Conjecture 1.1 has attracted considerable attention. The original paper [11] established it for several classes of graphs, including regular and bipartite graphs, and Abiad, de Lima, Desai, Guo, Hogben, and Madrid [1] proved it for several additional classes. Further work has examined structural properties and the asymmetry between the two square energies [12], as well as a finer conjecture for unicyclic graphs [20]. Substantial progress was also made on the general case: Zhang [24] established

$$\min\{s^+(G), s^-(G)\} \ge n - \gamma(G),$$

where $\gamma(G)$ denotes the domination number of G—in particular, $\min\{s^+, s^-\} \ge n/2$—while Akbari, Kumar, Mohar, and Pragada [2] simultaneously proved the linear bound

$$\min\{s^+(G), s^-(G)\} \ge \frac{3n}{4};$$

see also [4] for refinements concerning the positive square energy, and [8, 9] for a systematic study of the semidefinite and conic methods behind these bounds.

Our main result proves Conjecture 1.1 in full.

Theorem 1.2. *Every connected graph G on n vertices satisfies*

$$\min\{s^+(G), s^-(G)\} \ge n - 1.$$

Our proof is based on a general relaxation: since the Hadamard squares $A_\pm \circ A_\pm$ are doubly nonnegative, bounds on the square energies follow from inequalities valid over the entire doubly nonnegative cone. At the heart of this paper is one such inequality, for matrices indexed by the vertices of a connected graph (Theorem 2.1), which we believe to be of independent interest.

Beyond Conjecture 1.1 itself, Theorem 1.2 resolves several related conjectures from the literature. More precisely, it proves the disconnected version of the square-energy conjecture [11, Conjecture 2], Elphick's conjecture on adjacency inertia recorded in [24, Conjecture 7.4], and conjectured lower bounds for positive and negative p-energies from [3, Conjecture 2] and [13, Conjecture 5.4]. It also establishes unconditionally a Nordhaus–Gaddum-type inequality that Elphick and Aouchiche [10, Theorem 4] derived under Conjecture 1.1. These consequences are presented in Section 4. In the companion paper [19], the same relaxation yields a positive square-energy strengthening of Turán's theorem; the two proofs are otherwise independent.

1.1. **Notation.** For real symmetric matrices X and Y, write

$$\langle X, Y\rangle := \operatorname{tr}(XY), \qquad \|X\|_F := \sqrt{\langle X, X\rangle}.$$

We use $X \ge 0$ for entrywise nonnegativity; a real symmetric matrix X is *doubly nonnegative* if $X \succeq 0$ and $X \ge 0$ (see [6] for a textbook treatment).

Let $\mathbf{1}$ be the all-ones vector, set $J := \mathbf{1}\mathbf{1}^\top$, and write $X \circ Y$ for the Hadamard product of X and Y. Whenever a graph G is fixed and $u \in V(G)$, let e_u denote the corresponding standard basis vector; sums of the form $\sum_{uv \in E(G)}$ run over the edges of G, each edge counted once.

1.2. **Origin of the proof.** The proof given below is short, but its two ingredients—Theorem 2.1 and the reduction in Section 3—would be difficult to motivate in isolation. In this subsection, we give an informal explanation of where these ingredients come from. Readers interested only in the formal proof may skip directly to Section 2.

Our starting point is an observation underlying the semidefinite method of [5, 8, 9, 24]: the square energies are themselves optimal values of semidefinite programs. Indeed, for every $X \succeq 0$,

$$\langle A, X\rangle = \langle A_+, X\rangle - \langle A_-, X\rangle \le \langle A_+, X\rangle \le \|A_+\|_F \, \|X\|_F = \sqrt{s^+(G)}\; \|X\|_F \,,$$

with equality throughout at $X = A_+$, and therefore

$$s^+(G) = \max_{0\neq X\succeq 0} \frac{\max\{\langle A, X\rangle\,, 0\}^2}{\|X\|_F^2}, \qquad s^-(G) = \max_{0\neq X\succeq 0} \frac{\max\{-\langle A, X\rangle\,, 0\}^2}{\|X\|_F^2} \tag{2}$$

(equivalently, $s^{\pm}(G) = \min_{M\succeq 0} \|A \pm M\|_F^2$, the form emphasized in [24]). Consequently, an upper bound on both square energies is the same thing as a *certificate*: a constant β with

$$\langle A, X\rangle^2 \le \beta\; \|X\|_F^2 \qquad \text{for every } X \succeq 0.$$

Earlier certificates. Ando and Lin [5], proving a conjecture of Wocjan and Elphick [23], obtained the certificate $\beta = 2m\big(1 - \frac{1}{\chi(G)}\big)$, and Coutinho and Spier [8] sharpened it to $\beta = 2m\big(1 - \frac{1}{\chi_v(G)}\big)$, where $\chi_v(G) \le \chi(G)$ is the vector chromatic number. Both arguments use the same mechanism, with a parameter $t \in \{\chi(G), \chi_v(G)\}$. Applying the Cauchy–Schwarz inequality to the $2m$ nonzero entries of the adjacency matrix gives

$$\langle A, X\rangle^2 = \Big(\sum_{u,v} A_{uv}X_{uv}\Big)^2 \le 2m \sum_{u,v} A_{uv}X_{uv}^2 = 2m\; \langle A, X\circ X\rangle\,,$$

and the parameter t enters through an inequality on the doubly nonnegative cone: the matrix $Z := X \circ X$ satisfies $Z \ge 0$, and also $Z \succeq 0$ by the Schur product theorem [15, §7.5], and every such doubly nonnegative matrix obeys

$$\langle A, Z\rangle \le \left(1 - \frac{1}{t}\right) \langle J, Z\rangle\,. \tag{3}$$

Since $\langle J, X\circ X\rangle = \|X\|_F^2$, the two inequalities combine with (2) to give $s^{\pm}(G) \le 2m\big(1 - \frac{1}{t}\big)$. The conic optimization problem underlying (3) is studied systematically by Coutinho, Spier, and Zhang [9].

The idea behind (3). Following the original proofs in [5, 8], the key to (3) is the construction of a correlation matrix C (i.e., $C \succeq 0$ with $C_{vv} = 1$ for all $v \in V$) such that $C_{uv} \le -\frac{1}{t-1}$ for every edge $uv \in E$. For the parameter $t = \chi(G)$, such a C comes from a proper coloring made geometric: let $y_1, y_2, \ldots, y_t \in \mathbb{R}^{t-1}$ be the unit vectors of a regular simplex centered at the origin, so that $\langle y_i, y_j\rangle = -\frac{1}{t-1}$ for all $i \neq j$, and set $c_u := y_{\text{color of } u}$. The Gram matrix $C_{uv} := \langle c_u, c_v\rangle$ is then a correlation matrix with $C_{uv} = -\frac{1}{t-1}$ for all adjacent u and v, because adjacent vertices receive distinct colors. (Relaxing proper colorings to arbitrary unit vectors with $\langle c_u, c_v\rangle \le -\frac{1}{t-1}$ on edges—a *vector coloring*—is what sharpens the parameter to $t = \chi_v(G)$ [8].)

Under this construction $(J - C)\circ A \ge \frac{t}{t-1}A$, because $A \ge 0$ and $1 - C_{uv} \ge \frac{t}{t-1}$ on edges; hence $\langle (J-C)\circ A, Z\rangle \ge \frac{t}{t-1}\langle A, Z\rangle$ for every $Z \ge 0$. The claim (3) is then an instance of an identity valid for *every* correlation matrix C:

$$J - (J - C)\circ A \;=\; \underbrace{C}_{\succeq 0} \;+\; \underbrace{(J - C)\circ (J - A)}_{\ge 0}, \tag{4}$$

where $J - C \ge 0$ because $C \succeq 0$ with unit diagonal: every 2×2 principal minor of C gives $|C_{uv}| \le 1$. The right-hand side is a sum of a positive semidefinite and a nonnegative matrix, so pairing with any doubly nonnegative Z yields

$$\frac{t}{t-1}\,\langle A, Z\rangle \le \langle (J-C)\circ A, Z\rangle \le \langle J, Z\rangle\,,$$

the first inequality being the estimate just noted; this is exactly (3).

A weighted estimate. In the language of correlation matrices, the two estimates above collapse into a single Cauchy–Schwarz step, with weight $1 - C_{uv}$ on the edge uv:[1]

$$\langle A, X\rangle^2 = \Big(2\sum_{uv\in E} X_{uv}\Big)^2 \le \Phi_G(C)\cdot\langle (J-C)\circ A, X\circ X\rangle \le \Phi_G(C)\ \|X\|_F^2\,,$$

where the last inequality uses (4). For a correlation matrix C, define

$$\Phi_G(C) := \begin{cases} \displaystyle\sum_{uv\in E(G)} \frac{2}{1-C_{uv}}, & C_{uv} < 1 \text{ for every } uv \in E(G), \\ +\infty, & C_{uv} = 1 \text{ for some } uv \in E(G). \end{cases}$$

Here $C_{uv} \le 1$ for all u, v, since C is a correlation matrix. Every correlation matrix C with $\Phi_G(C) < \infty$ therefore gives a certificate of value $\Phi_G(C)$, and it is natural to minimize this quantity. Theorem 1.2 would follow from the bound

$$\min_C \Phi_G(C) \le q(G) := 2m - n + 1 \qquad \text{for every connected graph } G.$$

Bounding the optimum by duality. Minimizing Φ_G over correlation matrices is a convex problem that can be cast as a semidefinite program: the variable C ranges over the convex set

$$\mathrm{Corr}_n := \{C \in \mathbb{R}^{n\times n} : C = C^\top,\ C \succeq 0,\ C_{vv} = 1 \text{ for all } v\},$$

cut out by linear equations and one semidefinite constraint. The well-known first-order conditions for semidefinite programs (see [22, §3]) reveal what a minimizer C must look like: complementary slackness produces a doubly nonnegative matrix R, supported off-diagonal on $E(G)$, with

$$R_{uv} = \frac{1}{(1-C_{uv})^2} \quad (uv \in E), \qquad \sum_{uv\in E} \sqrt{R_{uv}} = \frac{\Phi_G(C)}{2}, \qquad \mathbf{1}^\top R\mathbf{1} = \Phi_G(C).$$

The two normalization identities suggest forgetting where R came from and, guided by homogeneity, guessing that

$$4\Big(\sum_{uv\in E} \sqrt{M_{uv}}\Big)^2 \le q(G)\cdot \mathbf{1}^\top M\mathbf{1} \tag{5}$$

for every connected graph G and every doubly nonnegative matrix M supported on the edges. Taking $M = R$ then gives $\Phi_G(C)^2 \le q(G)\,\Phi_G(C)$, i.e. $\Phi_G(C) \le q(G)$, as desired. Once formulated, the inequality admits a short inductive proof: it follows by induction on cut vertices—both q and Gram representations split there—with the 2-connected case handled by averaging the induction hypothesis over all one-vertex deletions.

A direct reduction. In the final write-up the semidefinite program disappears altogether. A folding trick disposes of the support hypothesis, so that (5) (Theorem 2.1) holds for all doubly nonnegative matrices, and applying it to $M = A_\pm \circ A_\pm$ yields Theorem 1.2 directly. In light of the variational characterization (2), this reduction says that Theorem 2.1 relaxes Theorem 1.2 from the Hadamard squares $\{X \circ X : X \succeq 0\}$, which compute the square energies exactly, to the full doubly nonnegative cone. The correlation-matrix program survives only in this overview, as the account of how the inequality was found.

[1] For the coloring matrix above, $1 - C_{uv} = \frac{t}{t-1}$ on every edge, so the weights are uniform; a general correlation matrix may price each edge differently.

1.3. **Organization.** The rest of the paper is organized accordingly: Section 2 proves Theorem 2.1, Section 3 deduces Theorem 1.2, and Section 4 collects consequences of the main theorem.

Formal verification. Theorem 1.2 has been formally verified in Lean 4 using Mathlib. The complete source code, documentation, reproducible build instructions, and kernel-axiom audit are available at `github.com/ShengtongZhang-alt/Sq`.

2. A doubly nonnegative matrix inequality

For a connected graph G, set

$$q(G) := 2|E(G)| - |V(G)| + 1;$$

by (1), Theorem 1.2 is equivalent to the upper bound $\max\{s^+(G), s^-(G)\} \le q(G)$, which we will deduce in Section 3 from the following inequality.

Theorem 2.1. *Let G be a simple connected graph and let M be a doubly nonnegative matrix indexed by $V(G)$. Then*

$$4 \cdot \Big(\sum_{uv \in E(G)} \sqrt{M_{uv}} \Big)^2 \le q(G) \cdot \mathbf{1}^\top M \mathbf{1}.$$

Proof. For any pair (G, M) as in the theorem, write

$$S(G,M) := \sum_{uv \in E(G)} \sqrt{M_{uv}}, \qquad T(M) := \mathbf{1}^\top M \mathbf{1},$$

so that the claim reads $4S(G,M)^2 \le q(G)\, T(M)$. We induct on $n := |V(G)|$. If $n = 1$, the result follows from $\mathbf{1}^\top M \mathbf{1} \ge 0$. If $n = 2$, then $G = K_2$, $q(K_2) = 1$, and $M = \left(\begin{smallmatrix} a & c \\ c & b \end{smallmatrix}\right)$ with $c \ge 0$; positive semidefiniteness gives $c \le \sqrt{ab} \le \frac{a+b}{2}$, so $4S(G,M)^2 = 4c \le a + b + 2c = T(M)$. Now assume $n \ge 3$ and that the theorem holds for all connected graphs on at most $n-1$ vertices.

We proceed in three steps: first we fold all non-edge entries of M onto the diagonal, so that we may assume $M_{uv} = 0$ whenever $u \ne v$ and $uv \notin E(G)$; then we treat the case where G has a cut vertex by splitting G into two smaller induced subgraphs; finally we handle the case where $G - v$ is connected for every vertex v by averaging the induction hypothesis over all vertex deletions.

Folding non-edges. Set

$$N := M + \sum_{\{u,v\} \in \binom{V(G)}{2} \setminus E(G)} M_{uv}\, (e_u - e_v)(e_u - e_v)^\top,$$

where each unordered non-edge $\{u, v\}$ is included exactly once. Each added term is positive semidefinite, cancels the corresponding non-edge entry of M, and moves its mass to the diagonal; hence $N \succeq 0$, $N \ge 0$, $N_{uv} = 0$ for non-edges $u \ne v$, and $N_{uv} = M_{uv}$ on edges, so the edge sums of M and N agree. Moreover $\mathbf{1}^\top (e_u - e_v)(e_u - e_v)^\top \mathbf{1} = (1-1)^2 = 0$, so $\mathbf{1}^\top N \mathbf{1} = \mathbf{1}^\top M \mathbf{1}$. Therefore, without loss of generality, we assume $M_{uv} = 0$ for non-edges $u \ne v$.

Cut-vertex case. Suppose $G - v$ is disconnected for some v. Group its connected components into two nonempty unions U, W and set

$$G_1 := G[U \cup \{v\}], \qquad G_2 := G[W \cup \{v\}].$$

Both are connected (each component of $G - v$ sends an edge to v), they share only v, their edge sets partition $E(G)$, and counting vertices and edges gives

$$q(G_1) + q(G_2) = q(G).$$

Since $M \succeq 0$, it admits a Gram representation: there are vectors z_u, $u \in V(G)$, in a Euclidean space, with norm $\|\cdot\|$, such that $M_{uv} = \langle z_u, z_v \rangle$ for all $u, v \in V(G)$. No edge joins U and W, so the support hypothesis makes the spans $\mathcal{U} := \operatorname{span}\{z_u : u \in U\}$ and $\mathcal{W} := \operatorname{span}\{z_w : w \in W\}$ orthogonal. Consider the orthogonal decomposition of z_v

$$z_v = p_U + p_W + p_0, \qquad p_U \in \mathcal{U}, \quad p_W \in \mathcal{W}, \quad p_0 \perp \mathcal{U} + \mathcal{W}.$$

Let M_1 be the Gram matrix of $(z_u)_{u \in U}$ together with $p_U + p_0$ at v, and M_2 that of $(z_w)_{w \in W}$ together with p_W at v. Every off-diagonal entry of M_1 or M_2 involving v agrees with the corresponding entry of M: for $u \in U$, $\langle z_u, p_U + p_0 \rangle = \langle z_u, z_v \rangle$ because $z_u \perp p_W, p_0$, and likewise on the other side. As all their other entries are entries of M or squared norms, M_1 and M_2 are entrywise nonnegative; being Gram matrices, they are also positive semidefinite. Thus both pairs (G_1, M_1) and (G_2, M_2) satisfy the hypotheses of the theorem, and the edge sums split,

$$S(G, M) = S(G_1, M_1) + S(G_2, M_2).$$

Moreover $T(M) = \left\| \sum_{u \in V(G)} z_u \right\|^2$ splits into the orthogonal parts $\sum_{u \in U} z_u + p_U$, $\sum_{w \in W} z_w + p_W$, and p_0, so

$$T(M) = T(M_1) + T(M_2).$$

By induction, $2S(G_i, M_i) \le \sqrt{q(G_i)\, T(M_i)}$, so the scalar Cauchy–Schwarz inequality gives

$$2S(G, M) \le \sqrt{q(G_1) T(M_1)} + \sqrt{q(G_2) T(M_2)} \le \sqrt{q(G)\, T(M)}.$$

No-cut-vertex case. Now $G - v$ is connected for every v, so the induction hypothesis can be averaged over all vertex deletions. Write $m := |E(G)|$, $q := q(G)$, and split $T(M)$ into diagonal and edge mass,

$$T(M) = d_0 + 2w, \qquad d_0 := \operatorname{tr} M, \quad w := \sum_{uv \in E(G)} M_{uv}.$$

Deleting a vertex v removes from $S(G, M)$ the terms of the edges at v:

$$S(G - v, M - v) = S(G, M) - \sigma_v, \qquad \sigma_v := \sum_{u \sim v} \sqrt{M_{uv}},$$

where $M - v$ denotes the submatrix of M obtained by deleting the row and column corresponding to v. Also, $G - v$ has $n - 1$ vertices and $m - d(v)$ edges, $d(v)$ being the degree of v, so $q(G - v) = q + 1 - 2d(v)$. The induction hypothesis for the pair $(G - v, M - v)$ therefore reads

$$2\big(S(G, M) - \sigma_v\big) \le \sqrt{q(G - v)\, T(M - v)}.$$

Sum over all v: every edge has two endpoints, so $\sum_v \sigma_v = 2S(G, M)$, and Cauchy–Schwarz gives

$$2(n - 2)\, S(G, M) \le \sum_v \sqrt{q(G - v)\, T(M - v)} \le \sqrt{\Big(\sum_v q(G - v)\Big)\Big(\sum_v T(M - v)\Big)}.$$

The two sums are elementary counts. Using $\sum_v d(v) = 2m$ and $4m = 2q + 2n - 2$, we have

$$\sum_v q(G - v) = \sum_v (q + 1 - 2d(v)) = n(q + 1) - 4m = (n - 2)(q - 1).$$

Since each diagonal entry of M survives in $n - 1$ of the submatrices $M - v$ while each edge entry survives in $n - 2$ of them, we also have

$$\sum_v T(M - v) = (n - 1)d_0 + 2(n - 2)w = (n - 2)\, T(M) + d_0.$$

Combining the last three displays and dividing by $(n-2)^2$ yields the *averaged estimate*

$$4S(G,M)^2 \le (q-1)\,T(M) + \frac{q-1}{n-2}\,d_0. \tag{6}$$

Cauchy–Schwarz over the m edges gives a second, *flat*, estimate:

$$4S(G,M)^2 \le 4m \sum_{uv\in E(G)} M_{uv} = 4mw. \tag{7}$$

Now we consider the following two cases, depending on the relative sizes of w and d_0.

Case 1. If $2(n-1)w \le qd_0$, then $4m = 2q + 2(n-1)$ turns (7) into

$$4S(G,M)^2 \le 4mw = 2qw + 2(n-1)w \le 2qw + qd_0 = q\,T(M).$$

Case 2. If instead $2(n-1)w > qd_0$, we use the averaged estimate (6), for which it suffices to show that $\frac{q-1}{n-2}\,d_0 \le T(M)$. Set $\beta := m - n + 1$, which is nonnegative because G is connected. Since $q - 1 = (n-2) + 2\beta$ and $T(M) = d_0 + 2w$, this reduces to

$$\beta\, d_0 \le (n-2)\, w.$$

Since G is simple, $2\beta = 2m-2n+2 \le (n-1)(n-2)$, so $q(n-2)-2\beta(n-1) = (n-1)(n-2)-2\beta \ge 0$. Multiplying the case hypothesis $qd_0 < 2(n-1)w$ by $n-2$,

$$2\beta(n-1)\,d_0 \le q(n-2)\,d_0 < 2(n-1)(n-2)\,w,$$

and dividing by $2(n-1)$ gives $\beta d_0 < (n-2)w$, as needed. Then (6) yields

$$4S(G,M)^2 \le (q-1)\,T(M) + T(M) = q\,T(M),$$

completing the induction. □

3. Proof of the main theorem

In this section, we directly deduce Theorem 1.2 from Theorem 2.1.

Proof of Theorem 1.2. The case $n = 1$ is trivial, so let $n \ge 2$; then G has an edge, and since $\operatorname{tr} A = 0$, the matrix A has eigenvalues of both signs, so $s^+(G) > 0$ and $s^-(G) > 0$.

Take $M := A_+ \circ A_+$, which satisfies $M \ge 0$ and, by the Schur product theorem, $M \succeq 0$. Its entries are $M_{uv} = (A_+)^2_{uv}$, so, using $A_+A_- = 0$,

$$\sum_{uv\in E(G)} \sqrt{M_{uv}} = \sum_{uv\in E(G)} \big|(A_+)_{uv}\big| \ge \sum_{uv\in E(G)} (A_+)_{uv} = \tfrac{1}{2}\,\langle A, A_+\rangle = \tfrac{1}{2}\operatorname{tr}(A_+^2) = \tfrac{1}{2}\,s^+(G),$$

while

$$\mathbf{1}^\top M \mathbf{1} = \sum_{u,v} (A_+)^2_{uv} = \|A_+\|_F^2 = s^+(G).$$

Theorem 2.1 therefore gives

$$s^+(G)^2 \le 4\Big(\sum_{uv\in E(G)} \sqrt{M_{uv}}\Big)^2 \le q(G)\,s^+(G),$$

and dividing by $s^+(G) > 0$ yields $s^+(G) \le q(G)$.

For the negative part, take $M := A_- \circ A_-$. Now $\langle A, A_-\rangle = -\operatorname{tr}(A_-^2) = -s^-(G)$, so, using $|x| \ge -x$,

$$\sum_{uv\in E(G)} \sqrt{M_{uv}} = \sum_{uv\in E(G)} \big|(A_-)_{uv}\big| \ge -\sum_{uv\in E(G)} (A_-)_{uv} = \tfrac{1}{2}\,s^-(G),$$

and $\mathbf{1}^\top M \mathbf{1} = \|A_-\|_F^2 = s^-(G)$ as before, whence $s^-(G) \le q(G)$. By (1),

$$s^\pm(G) = 2m - s^\mp(G) \ge 2m - q(G) = n - 1.$$

□

4. Consequences of the main theorem

We record several consequences of Theorem 1.2. Throughout this section, $\kappa(G)$ denotes the number of connected components of G, and

$$c(G) := |E(G)| - |V(G)| + \kappa(G)$$

denotes its cyclomatic number.

4.1. **Disconnected graphs.** The following corollary proves the disconnected version of the square-energy conjecture [11, Conjecture 2].

Corollary 4.1. *Let G be a graph on n vertices with $\kappa(G)$ connected components. Then*

$$\min\{s^+(G), s^-(G)\} \geq n - \kappa(G).$$

Proof. Let $G_1, \ldots, G_{\kappa(G)}$ be the connected components of G, and set $n_i := |V(G_i)|$. Since the adjacency matrix of G is block diagonal, s^+ and s^- are additive over connected components. Therefore Theorem 1.2 gives, for $\sigma \in \{+, -\}$,

$$s^\sigma(G) = \sum_{i=1}^{\kappa(G)} s^\sigma(G_i) \geq \sum_{i=1}^{\kappa(G)} (n_i - 1) = n - \kappa(G). \qquad \square$$

4.2. **Asymmetry of the square energies.** Elphick and Linz [12] call $s^+(G) - s^-(G)$ the *squared spread* of G. Combining Corollary 4.1 with (1) gives, for either sign σ,

$$n - \kappa(G) \leq s^\sigma(G) \leq 2m - n + \kappa(G), \qquad \text{that is,} \qquad |s^\sigma(G) - m| \leq c(G).$$

It follows that

$$|s^+(G) - s^-(G)| \leq 2c(G). \tag{8}$$

The next result strengthens and makes unconditional the upper bound in [12, Proposition 4.1].

Corollary 4.2. *Every graph G on n vertices satisfies*

$$|s^+(G) - s^-(G)| \leq (n-1)(n-2).$$

For $n \geq 3$, equality holds if and only if $G \cong K_n$.

Proof. Let $\kappa := \kappa(G)$. If the connected components of G have orders $n_1, \ldots, n_\kappa$, then

$$|E(G)| \leq \sum_{i=1}^{\kappa} \binom{n_i}{2} \leq \binom{n-\kappa+1}{2}.$$

The second inequality is attained only when one component has $n - \kappa + 1$ vertices and all the remaining components are isolated vertices. It follows that

$$c(G) = |E(G)| - n + \kappa \leq \binom{n-\kappa}{2} \leq \binom{n-1}{2}.$$

By (8), we obtain

$$|s^+(G) - s^-(G)| \leq 2c(G) \leq 2\binom{n-1}{2} = (n-1)(n-2).$$

Suppose that $n \geq 3$ and equality holds. Then equality must hold in

$$c(G) \leq \binom{n-\kappa}{2} \leq \binom{n-1}{2},$$

which forces $\kappa = 1$ and $|E(G)| = \binom{n}{2}$. Hence $G \cong K_n$. Conversely, the adjacency eigenvalues of K_n are $n-1$ and -1 with multiplicity $n-1$, so

$$s^+(K_n) - s^-(K_n) = (n-1)^2 - (n-1) = (n-1)(n-2). \qquad \square$$

4.3. **Adjacency inertia.** For a graph G, let $n^+(G)$, $n^0(G)$, and $n^-(G)$ denote, respectively, the numbers of positive, zero, and negative eigenvalues of its adjacency matrix, counted with multiplicity. Let $\iota(G)$ denote the number of isolated vertices.

Zhang [24, Corollary 1.5] proved that every graph without isolated vertices satisfies

$$\min\{s^+(G), s^-(G)\} \geq \max\{n^+(G), n^0(G), n^-(G)\}.$$

The following result proves the stronger conjecture of Elphick recorded in [24, Conjecture 7.4] and extends it to graphs with isolated vertices.

Corollary 4.3. *Every graph G satisfies*

$$\min\{s^+(G), s^-(G)\} \geq n^0(G) - \iota(G) + \max\{n^+(G), n^-(G)\}.$$

Proof. Let $\eta := \kappa(G) - \iota(G)$ denote the number of connected components of G with at least two vertices. The adjacency matrix of any such component has at least one positive and at least one negative eigenvalue. Hence

$$\min\{n^+(G), n^-(G)\} \geq \eta.$$

Since $n = n^+(G) + n^0(G) + n^-(G)$, we obtain

$$\begin{aligned} n^0(G) - \iota(G) + \max\{n^+(G), n^-(G)\} &= n - \iota(G) - \min\{n^+(G), n^-(G)\} \\ &\leq n - \iota(G) - \eta \\ &= n - \kappa(G). \end{aligned}$$

The result follows from Corollary 4.1. □

4.4. **Positive and negative p-energies.** For $p > 0$, the corresponding positive and negative p-energies are defined by

$$\mathcal{E}_p^+(G) := \sum_{\lambda_i > 0} |\lambda_i|^p, \qquad \mathcal{E}_p^-(G) := \sum_{\lambda_i < 0} |\lambda_i|^p.$$

These quantities have also been studied in several recent papers [3, 7, 13, 18, 21]. Our main theorem resolves a number of conjectures surrounding these energies.

For connected graphs, the next result proves [13, Conjecture 5.4]. Its negative-sign case for $p \geq 2$ also proves [3, Conjecture 2].

Corollary 4.4. *Let G be a graph on n vertices. For every $p > 0$ and $\sigma \in \{+, -\}$,*

$$\mathcal{E}_p^\sigma(G) \geq \begin{cases} (n - \kappa(G))^{p/2}, & 0 < p \leq 2, \\ n - \kappa(G), & p \geq 2. \end{cases}$$

Proof. Set $N := n - \kappa(G)$. The case $N = 0$ is immediate. Suppose that $N > 0$, fix $\sigma \in \{+, -\}$, and let $x_1, \ldots, x_r$ be the absolute values of the adjacency eigenvalues of sign σ. By Corollary 4.1,

$$\sum_{i=1}^{r} x_i^2 = s^\sigma(G) \geq N.$$

If $0 < p \leq 2$, then

$$\mathcal{E}_p^\sigma(G) = \sum_{i=1}^{r} x_i^p \geq \left(\sum_{i=1}^{r} x_i^2 \right)^{p/2} \geq N^{p/2}.$$

Now let $p \geq 2$. By Hölder's inequality,

$$\mathcal{E}_p^\sigma(G) \geq r^{1-p/2} \left(\sum_{i=1}^{r} x_i^2 \right)^{p/2}.$$

Every nontrivial connected component of G contributes an adjacency eigenvalue of sign opposite to σ, while every isolated vertex contributes a zero eigenvalue. Thus at least $\kappa(G)$ eigenvalues are not among $x_1, \dots, x_r$, and hence $r \le n - \kappa(G) = N$. Since $1 - p/2 \le 0$, it follows that

$$\mathcal{E}_p^{\sigma}(G) \ge N^{1-p/2} N^{p/2} = N. \qquad \square$$

For a connected graph, Corollary 4.4 gives

$$\min\{\mathcal{E}_p^{+}(G), \mathcal{E}_p^{-}(G)\} \ge (n-1)^{p/2} \qquad (0 < p \le 2).$$

This proves [13, Conjecture 5.4]; the range $0 < p < 1$ was previously proved in [13, Theorem 5.7]. With the convention $\mathcal{E}_0^{\pm}(G) := n^{\pm}(G)$, the corresponding assertion at $p = 0$ is immediate.

If G is connected and $p \ge 2$, then the negative-sign bound gives

$$\mathcal{E}_p^{-}(G) \ge n - 1 = \mathcal{E}_p^{-}(K_n).$$

This proves [3, Conjecture 2]. The conjecture was previously known for all real $p \ge 4$ by [3] and for every integer $p \ge 3$ by [7].

The bounds in Corollary 4.4 are sharp. For $0 < p < 2$, equality is attained by the disjoint union of a star and isolated vertices. For $p = 2$, equality holds for every forest. For $p > 2$, equality is attained by a disjoint union of edges and isolated vertices. In the connected negative-sign case, equality is attained by K_n.

For connected graphs and $p > 2$, Corollary 4.4 gives

$$\mathcal{E}_p^{+}(G) \ge n - 1.$$

The stronger conjecture

$$\mathcal{E}_p^{+}(G) \ge \mathcal{E}_p^{+}(P_n),$$

where P_n is the path on n vertices, remains open in general. It is known for connected bipartite graphs for every real $p \ge 2$, for all connected graphs when $p \ge 3$ is an odd integer, and for all connected graphs when $p = 4$; see [18].

4.5. **Graph complements.** Let $\overline{G}$ denote the complement of G. Elphick and Aouchiche [10, Conjecture 6] conjectured that

$$s^{+}(G) + s^{+}(\overline{G}) \le (n-1)^2$$

and observed that this inequality would follow from Conjecture 1.1. The following corollary proves their conjecture and its negative-sign analogue.

Corollary 4.5. *Let G be a graph on $n \ge 2$ vertices. For every $\sigma \in \{+, -\}$,*

$$s^{\sigma}(G) + s^{\sigma}(\overline{G}) \le (n-1)^2.$$

For $\sigma = +$, equality holds if and only if G is complete or edgeless. For $\sigma = -$, equality holds if and only if $n = 2$.

Proof. Fix $\sigma \in \{+, -\}$, and let τ denote the opposite sign. Write $\kappa := \kappa(G)$ and $\overline{\kappa} := \kappa(\overline{G})$. By Corollary 4.1,

$$s^{\tau}(G) + s^{\tau}(\overline{G}) \ge 2n - \kappa - \overline{\kappa}.$$

Since

$$s^{+}(G) + s^{-}(G) + s^{+}(\overline{G}) + s^{-}(\overline{G}) = n(n-1),$$

we obtain

$$s^{\sigma}(G) + s^{\sigma}(\overline{G}) \le n(n-1) - 2n + \kappa + \overline{\kappa}.$$

At least one of G and $\overline{G}$ is connected, and hence $\kappa + \overline{\kappa} \le n + 1$. Therefore

$$s^{\sigma}(G) + s^{\sigma}(\overline{G}) \le (n-1)^2.$$

Suppose that equality holds. Then $\kappa+\overline{\kappa}=n+1$. Since at least one of G and $\overline{G}$ is connected, the other has n connected components. Thus one of the two graphs is complete and the other is edgeless. Now

$$s^+(K_n)=(n-1)^2, \qquad s^-(K_n)=n-1,$$

whereas both square energies of an edgeless graph vanish. The equality statements follow. □

Acknowledgments and AI disclosure

We used ChatGPT to generate exploratory code for testing conjectures and to help polish the language of proof drafts. All mathematical content was verified by the authors, and the proof of Theorem 1.2 has additionally been formally verified in the Lean proof assistant; the formalization is available at https://github.com/ShengtongZhang-alt/Sq. The core ideas and the overall strategy of the proof were developed independently by the authors; the motivation and the path leading to the argument are described in the overview; any suggestions from the tool were modified or discarded as needed, and the authors take full responsibility for the correctness and originality of the paper.

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Institute for Interdisciplinary Information Sciences, Tsinghua University, Beijing 100084, P. R. China
Email address: liuyinch23@mails.tsinghua.edu.cn

School of Mathematical Sciences, University of Science and Technology of China, Hefei 230026, P. R. China
Email address: tangquanyu827@gmail.com

Department of Mathematics, Stanford University, Stanford, CA 94305, USA
Email address: stzh1555@stanford.edu